

Library Management System V2 - Project Report

About Me

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Subject: Modern Application Development – II Project

Course: Diploma in Programming

About The Project

This is an online Library Management System web application project with Role Based Access Control (RBAC). In this project there will be an Admin (Librarian) and several users. There are many Sections and each section can have multiple books, the librarian can perform CRUD operations (via APIs) on Sections and Books. A new user can register on Library Management System website and then he/she can start reading books after borrowing. User can request admin to issue a book. The admin has authority to either grant or reject the request. A user can get maximum 5 books issued. Both librarian and user can search and get the search results based on Book, Section and/or Author. Both librarian and user can see some basic stats. User can also buy a book.

APIs for performing CRUD on Sections & Books

1. ["/api/sections"](#) (for Sections)
2. ["/api/books"](#) (for Books)

Note: For running Library Management System web application, please refer to [readme.md](#) file present inside the root folder of the project. The database schema has already been created with Admin login credentials mentioned below. Few sections with some books have also been created.

Admin Credentials

<u>Fields</u>	<u>Admin</u>	<u>Ravi Kumar (user)</u>
Email	admin@gmail.com	user1@gmail.com
Password	Pass@123	Pass@123

Packages/Libraries Used in the Project

- **Flask** (framework for creating this Library Management System web application)
- **Vue2** (for frontend)
- **Flask SQLAlchemy** (for working with database)
- **Flask Security Too** (for security and RBAC)
- **Flask RESTful** (for API)
- **fpdf** (for pdf generation and download)
- **Flask Caching** (for caching)
- **Celery and Redis** (for backend jobs)
- **MailHog** (for testing periodic emails)

Key Features

- Role Based Access Control (RBAC)
- CRUD on Sections & Books (via APIs) – for **admin** role
- Request Book, Return Book, Buy Book (by paying a price), visiting all sections & books – for **user** role
- User can rate and provide feedback/review for a book
- Search functionality based on book name, section name, authors – for both **admin** & **user**
- User can have maximum 5 books issued and/or requested at once
- Admin can revoke/take back an issued book from a user if he/she does not return the book on time
- Backend Jobs:
 - **Daily Reminder** (via email) – If user keeps book for more than 6 days
 - **Monthly Activity Report** (via email) – Admin gets an activity report on day one of every month
 - **CSV Report** – Admin can download a CSV report containing activities from beginning to end

DB Schema Design

